BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

PHYSICS — SSC-I (9th Grade)

Syllabus: Unit 5 — Pressure and Deformation in Solids

Total Marks: 65 | **Time Allowed: 2 hours 30 minutes** | **Version: 1**

General Instructions:

1. This paper has **three sections**: A (MCQs), B (Short Questions), C (Long Questions).
2. **Section A** (12 marks): 12 MCQs. Mark the correct bubble. Time: 15 minutes.
3. **Section B** (33 marks): 11 short questions × 3 marks each. Answer *either* the main question *or* the OR alternative.
4. **Section C** (20 marks): 4 long questions × 5 marks each. Answer *either* the main question *or* the OR alternative in B and C.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks) [15 min]

Q.1: Choose the correct answer and fill in the corresponding bubble.

#	Question	A	B	C	D	Answer
1	The ability of a deformed body to return to its original shape and size when the deforming force is removed is called:	Plasticity	Elasticity	Rigidity	Ductility	Ⓐ Ⓑ Ⓒ Ⓓ
2	Hooke's law holds good up to the:	Yield limit	Breaking point	Proportional limit	Plastic limit	Ⓐ Ⓑ Ⓒ Ⓓ
3	The SI unit of the spring constant is:	N/m	N·m	kg/m	Pa	Ⓐ Ⓑ Ⓒ Ⓓ
4	The SI unit of pressure is:	bar	newton	psi	pascal	Ⓐ Ⓑ Ⓒ Ⓓ
5	A mass of 2 kg is hung on a spring which displaces it by 5 cm. The spring constant is:	400 N/m	40 N/m	4 N/m	4000 N/m	Ⓐ Ⓑ Ⓒ Ⓓ
6	The most elastic material among the following is:	Rubber	Wood	Glass	Steel	Ⓐ Ⓑ Ⓒ Ⓓ
7	Atmospheric pressure at sea level is approximately:	1.013×10^3 Pa	1.013×10^5 Pa	760 Pa	101 Pa	Ⓐ Ⓑ Ⓒ Ⓓ
8	Pressure in a liquid at depth h is given by:	$P = mgh$	$P = \rho g/h$	$P = \rho gh$	$P = mg/A$	Ⓐ Ⓑ Ⓒ Ⓓ
9	Atmospheric pressure is commonly measured using:	Hygrometer	Barometer	Manometer	Thermometer	Ⓐ Ⓑ Ⓒ Ⓓ
10	Pascal's principle states that an external pressure applied to an enclosed fluid is transmitted:	Only downward	Only to the walls	Unchanged to every point	With increasing magnitude	Ⓐ Ⓑ Ⓒ Ⓓ
11	The atmospheric pressure will be smallest at:	Islamabad	Karachi	Lahore	Murree	Ⓐ Ⓑ Ⓒ Ⓓ
12	In a hydraulic lift, if piston 2 area is 100 times piston 1 area, the output force is:	100 times input force	Same as input force	1/100 of input force	10 times input force	Ⓐ Ⓑ Ⓒ Ⓓ

SECTION B — Short Answer Questions (11 × 3 = 33 Marks)

Q.2: Attempt **all** questions. Answer *either* the main question *or* the OR alternative.

Part	Question	Marks	OR	OR Question	Marks
(i)	Define **elasticity** and **elastic limit**. Give one example of an elastic material and one inelastic material.	**3**	**OR**	What are **inelastic materials**? Give three examples. Why do they not return to their original shape after the deforming force is removed?	**3**
(ii)	State **Hooke's Law**. Write its mathematical form and define the spring constant with its SI unit.	**3**	**OR**	Draw and label a **force-extension graph** for an elastic solid. Identify the limit of proportionality and explain what happens beyond it.	**3**
(iii)	A spring chair compresses by 8 cm when a person of mass 60 kg sits on it. Calculate the **spring constant** of the chair spring. ($g = 10\ m/s^2$)	**3**	**OR**	Name and briefly describe **three applications of Hooke's Law** in everyday instruments.	**3**
(iv)	Define **pressure**. Write its formula and SI unit. Explain why a sharp knife cuts better than a blunt knife using the concept of pressure.	**3**	**OR**	A person of mass 50 kg stands on one foot of area $0.02\ m^2$. Calculate the **pressure** exerted on the ground. ($g = 9.8\ m/s^2$)	**3**
(v)	Define **atmospheric pressure**. Why do we not feel the enormous weight of the atmosphere pushing down on us?	**3**	**OR**	Explain with two examples how **atmospheric pressure** is used in everyday life (drinking through a straw and drawing liquid by syringe).	**3**
(vi)	Describe the construction and working of a **mercury barometer**. What does the height of the mercury column represent?	**3**	**OR**	Explain how **atmospheric pressure varies with altitude**. Why is it difficult to cook food at high altitudes?	**3**
(vii)	Show that the pressure exerted by a liquid column is given by $\mathbf{P = \rho g h}$. What two factors does liquid pressure depend on?	**3**	**OR**	A submarine dives to a depth of 500 m in sea water of density $1025\ kg/m^3$. Calculate the **pressure** on the submarine. ($g = 9.8\ m/s^2$)	**3**
(viii)	Define a **manometer**. State its working principle and write the formula used to find fluid pressure with it.	**3**	**OR**	Give **four applications** of a manometer in industry and laboratories.	**3**
(ix)	State **Pascal's Principle**. Explain with the bicycle tyre example how pressure is transmitted equally in all directions.	**3**	**OR**	Differentiate between a **manometer** and a **barometer**. Give two differences in their construction and use.	**3**
(x)	Explain the working of a **hydraulic lift**. Write the equation $F_2 = (A_2/A_1)\ F_1$ and explain how force is multiplied.	**3**	**OR**	Explain the **hydraulic car brake system** step by step: brake pedal → master cylinder → brake lines → brake calipers.	**3**
(xi)	Why do **static fluids** always exert a force perpendicular to a surface? Explain with reference to Pascal's Principle.	**3**	**OR**	Explain why **divers wear special suits** in deep water. What would happen if a diver descends without protection to 8.5 km depth?	**3**

SECTION C — Long Questions (4 × 5 = 20 Marks)

Q.3: Attempt **all** questions. Answer *either* the main question *or* the OR alternative.

Q.	Question	Marks	OR	OR Question	Marks
3(i)	Define **elasticity** and **elastic limit**. State Hooke's Law with its mathematical form. Define spring constant, state its unit and derive it. Draw a fully labelled force-extension graph showing: proportional limit, elastic limit, and breaking point.	**5**	**OR**	Explain the **stress-strain curve** for a metal. Define and identify on your diagram: (a) proportional limit A, (b) elastic limit B, (c) ultimate stress point C, (d) breaking point D. Distinguish between the elastic region and plastic region.	**5**
3(ii)	Define **pressure** and write P = F/A. Explain the effect of area on pressure with three real-life examples (thumb pin, high-heel shoes, sharp knife). A girl of mass 50 kg wears heels of area 2 cm^2 in contact with the ground. Calculate the pressure she exerts. ($g = 9.8\ m/s^2$)	**5**	**OR**	Explain **atmospheric pressure**: define it, state its value in Pa, atm, and mmHg. Describe the construction and working of a **mercury barometer**. How does pressure vary with altitude and what effect does this have on weather?	**5**
3(iii)	Derive the formula **P = ρgh** for liquid pressure. Explain how pressure varies with depth using the drill-hole-in-can activity. A submarine is at 8.5 km depth in sea water ($\rho = 1000\ kg/m^3$). Calculate the pressure on it. ($g = 9.8\ m/s^2$)	**5**	**OR**	Define a **manometer** and state its formula $P = \rho gh$. Draw and label the three types of manometer (U-tube, well-reservoir, inclined). List **five applications** of a manometer. How does it differ from a barometer?	**5**
3(iv)	State **Pascal's Principle**. Derive the hydraulic lift equation $F_2 = (A_2/A_1)\ F_1$ and explain the concept of **force multiplier**. A hydraulic lift has narrow cylinder area 0.002 m^2 and wide cylinder area 0.9 m^2. Find the input force needed to lift a car of mass 1800 kg. ($g = 9.8\ m/s^2$)	**5**	**OR**	Explain the **hydraulic car brake system** with a labelled diagram. Name all four components (brake pedal, master cylinder, brake lines, brake calipers). How does Pascal's Principle explain equal braking force on all wheels? Why is it dangerous if air enters the brake fluid?	**5**

Invigilator's Signature: ____________________ Candidate's Signature: ____________________

Chapters Covered: Unit 5 — Pressure and Deformation in Solids

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